SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	**CSE AI**	**Date:**	**13 / 10 / 2026**

Code of Conduct

I ***__k Deepthi / 192472211_____*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____

Questions

Q1. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Given Values:

- Total channels = 560
- Cluster size = 7

Formula:

$$\text{Channels per cell} = \frac{\text{Total channels}}{\text{Cluster size}}$$

Calculation:

$$Channels\ per\ cell = \frac{560}{7} = 80$$

: **80 channels per cell.**

Q2. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Given Values:

Number of carriers = 100

Time slots per carrier = 8

Formula: Total channels = Carriers × Time slots

Calculation:

= 100 × 8

= 800

Answer: 800 simultaneous voice channels.

Q3. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Given Values:

Carrier power = −60 dBm

Interference power = −92 dBm

Formula: C/I (dB) = Carrier power − Interference power

Calculation:

= −60 − (−92)

= 32 dB

Answer: 32 dB.

Q4. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Given Values:

Number of cells = 4

Channels per cell = 15

Formula: Total channels = Number of cells × Channels per cell

Calculation:

= 4 × 15

= 60

Answer: 60 channels.

Q5. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Given Values:

Frame duration = 5 ms

Number of slots = 8

Formula: Slot duration = Frame duration ÷ Number of slots

Calculation:

= 5 ÷ 8

= 0.625 ms

Answer: 0.625 ms per slot.

Q6. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Given Values:

Total bandwidth = 24 MHz = 24,000 kHz

Carrier spacing = 200 kHz

Formula: Number of carriers = Total bandwidth ÷ Carrier spacing

Calculation:

= 24,000 ÷ 200

= 120

Answer: 120 carriers.

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Q7. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Given Values:

Number of cells = 6

Calls per cell = 250

Formula: Capacity = Number of cells × Calls per cell

Calculation:

= 6 × 250

= 1,500

Answer: 1,500 simultaneous calls.

Q8. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Given Values:

Number of channels = 150

Bandwidth per channel = 25 kHz

Formula: Total bandwidth = Number of channels × Bandwidth per channel

Calculation:

= 150 × 25

= 3,750 kHz ÷ 1,000

= 3.75 MHz

Answer: 3.75 MHz.

Q9. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Given Values:
Regional centers = 8
Base stations per center = 40
Users per base station = 250
Formula: Total capacity = Regional centers × Base stations per center × Users per base station
Calculation:
= 8 × 40 × 250
= 80,000
Answer: 80,000 users.

Q10. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Given Values:

Total bandwidth = 6 MHz = 6,000 kHz

Bandwidth per channel = 30 kHz

Formula: Number of channels = Total bandwidth ÷ Bandwidth per channel

Calculation:

= 6,000 ÷ 30

= 200

Answer: 200 channels.

Q11. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Given Values:

Subscribers = 1,200

Calls per subscriber per hour = 4

Average call duration = 2.5 minutes

Formula: Traffic (Erlangs) = Subscribers × Calls per hour × Call duration (minutes) ÷ 60

Calculation:

= 1,200 × 4 × 2.5 ÷ 60

= 12,000 ÷ 60

= 200 Erlangs

Answer: 200 Erlangs.

Q12. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Given Values:

Base stations = 1,200

Users per base station = 600

Formula: Total capacity = Base stations × Users per base station

Calculation:

= 1,200 × 600

= 720,000

Answer: 720,000 users.

Q13. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Given Values:

Maximum capacity = 12,000 calls/s

Actual load = 9,000 calls/s

Formula: Utilization (%) = (Actual load ÷ Maximum capacity) × 100

Calculation:

= (9,000 ÷ 12,000) × 100

= 75%

Answer: 75% utilization.

Q14. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Given Values:

Total bandwidth = 30 MHz = 30,000 kHz

Channel spacing = 200 kHz

Formula: Number of channels = Total bandwidth ÷ Channel spacing

Calculation:

= 30,000 ÷ 200

= 150

Answer: 150 channels.

Q15. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Given Values:

Total handoff attempts = 2,500

Success rate = 99%

Formula: Successful handoffs = (Success rate × Total attempts) ÷ 100

Calculation:

= (99 × 2,500) ÷ 100

= 2,475

Answer: 2,475 successful handoffs.

Q16. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Given Values:

Number of cells = 7

Users per cell = 900

Formula: Total capacity = Number of cells × Users per cell

Calculation:

= 7 × 900

= 6,300

Answer: 6,300 users.

Q17. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Given Values:

Frame duration = 4.8 ms

Number of slots = 8

Formula: Slot duration = Frame duration ÷ Number of slots

Calculation:

= 4.8 ÷ 8

= 0.6 ms

Answer: 0.6 ms per slot.

Q18. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Given Values:

Number of cells = 9

Users per cell = 600

Formula: Total capacity = Number of cells × Users per cell

Calculation:

= 9 × 600

= 5,400

Answer: 5,400 users.

Q19. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Given Values:

Total calls per second = 60,000

Drop rate = 0.4%

Formula: Dropped calls per second = (Drop rate × Total calls per second) ÷ 100

Calculation:

= (0.4 × 60,000) ÷ 100

= 240

Answer: 240 dropped calls per second.

Q20. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

Given Values:

Radius = 4 km

$\pi = 3.14$

Formula: Area of a circle = πr^2

Calculation:

= 3.14×4^2

= 3.14×16

= 50.24 km²

Answer: Approximately 50.24 km².

Q21. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Given Values:

Total subscribers = 150,000

Users per cycle = 6,000

Formula: Paging cycles = Total subscribers ÷ Users per cycle

Calculation:

= 150,000 ÷ 6,000

= 25

Answer: 25 paging cycles.

Q22. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Given Values:

Signal power = −85 dBm

Noise power = −115 dBm

Formula: SNR (dB) = Signal power − Noise power

Calculation:

= −85 − (−115)

= 30 dB

Answer: 30 dB.

Q23. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Given Values:

Subscribers = 1,500

Calls per subscriber per hour = 4

Average call duration = 2.5 minutes

Formula: Traffic (Erlangs) = Subscribers × Calls per hour × Call duration (minutes) ÷ 60

Calculation:

= 1,500 × 4 × 2.5 ÷ 60

= 15,000 ÷ 60

= 250 Erlangs

Answer: 250 Erlangs.

Q24. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Given Values:

Number of control channels = 250

Bandwidth per channel = 12 kHz

Formula: Total bandwidth = Number of channels × Bandwidth per channel

Calculation:

= 250 × 12

= 3,000 kHz ÷ 1,000

= 3 MHz

Answer: 3 MHz.

Q25. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Given Values:

Simultaneous connections = 1,500

Data rate per connection = 600 kbps

Formula: Total throughput = Number of connections × Data rate per connection

Calculation:

= 1,500 × 600

= 900,000 kbps ÷ 1,000

= 900 Mbps

Answer: 900 Mbps.